Abbreviations

cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
oe or equivalent
SC Special Case
nfww not from wrong working
soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	Reflection $y = -1$	**2**	**B1** for each
1(a)(ii)	Triangle at (0, –3), (4, –1), (4, –3)	**2**	**B1** for translation $\begin{pmatrix} -2 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -5 \end{pmatrix}$ or for three correct vertices
1(a)(iii)	Triangle at (–2, 2), (–2, 6), (–4, 6)	**2**	**B1** for rotation about (0, 0) 90° clockwise or 90° anticlockwise with wrong centre or for three correct vertices
1(a)(iv)	Triangle at (–3, –1), (–3, –2), (–1, –1)	**2**	**B1** for scale factor $-\frac{1}{2}$ with wrong centre or scale factor $\frac{1}{2}$ with centre (0, 0) or for three correct vertices
1(b)(i)	$\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ cao	**1**	
1(b)(ii)	4.47 or 4.472…	**2**	**M1** for $(\textit{their } 2)^2 + (\textit{their } 4)^2$
1(b)(iii)	(7, 10)	**2**	**B1** for each
1(b)(iv)	$y = 2x - 4$ oe	**3**	**M1** for gradient = $\frac{6-2}{5-3}$ oe or answer $y = mx - 4$ **M1** for substituting (3, 2) or (5, 6) into $y = \textit{their } mx + c$ or into $y - k = \textit{their } m(x - h)$ or into $\textit{their } y = mx - 4$
1(b)(v)	(0, –4)	**1**	**FT** *their* **(b)(iv)**
2(a)(i)	$\frac{240}{(23+25)} \times 23$	**M1**	
2(a)(ii)	11 : 10	**2**	**M1** for 110 : 100 or better or **SC1** for 10 : 11, following boys 100, girls 110

Question	Answer	Marks	Partial Marks
2(a)(iii)	276	**2**	**M1** for $240 \times \left(1+\frac{15}{100}\right)$ oe or **B1** for 36 seen
2(a)(iv)	150	**3**	**M2** for $\frac{240}{100+60}$ [× 100] oe or **M1** for evidence of 160[%] associated 240
2(b)	464 000	**3**	**M1** for $256\,000 \times \left(1+\frac{2}{100}\right)^{30}$ oe **A1** for 463 700 to 463 710 **B1** for *their* more accurate answer seen and rounded to nearest 1000
2(c)	4.5[0]	**3**	**M2** for [x =] $\sqrt[32]{4.09}$ oe or **M1** for $(x)^{32} = 4.09$ oe If 0 scored, **SC2** for answer 3.6 or 3.59 or 3.588… or **SC1** for $\sqrt[32]{3.09}$ or 1.0358 to 1.036 seen
3(a)(i)	427 or 427.2 to 427.3…	**2**	**M1** for $\pi \times 8 \times 17$
3(a)(ii)	1010 or 1005….	**4**	**M2** for $\sqrt{17^2 - 8^2}$ oe or **M1** for $h^2 + 8^2 = 17^2$ oe **M1** for $\frac{1}{3} \times \pi \times 8^2 \times$ *their h* oe
3(a)(iii)	804 or 804.2 to 804.4 or 808	**1**	**FT** *their* **(ii)** × 0.8
3(a)(iv)	396 or 395.6 to 395.8 or 392	**1**	**FT** 1200 – *their* **(iii)**
3(b)(i)	$\frac{1}{54}$	**4**	**B3** for $\frac{\frac{4}{3}\pi r^3}{72\pi r^3}$ or better or **M2** for $\frac{\frac{4}{3} \times \pi \times r^3}{\pi \times (3r)^2 \times 8r}$ or $72 \times \pi \times r^3$ or **M1** for $\pi \times (3r)^2 \times 8r$ If 0 scored, **SC2** for answer of $\frac{1}{18}$
3(b)(ii)	972π final answer	**4**	**B2** for $r = \frac{9}{2}$ oe or **M1** for $4\pi r^2 = 81\pi$ or better **M1** for $2 \times \pi \times (3 \times$ *their r*$) \times (8 \times$ *their r*$)$ isw
4(a)	–1, 3	**2**	**B1** for each